

Applications of Algebra (Part II)

Name
Date
Course

Simple Interest Formula

$$A = P + Prt$$

A = Amount of money possessed after t years. P = Initial amount of money (i.e. principal), r = Interest rate (in decimal form).

Compound Interest Formula

$$A = P(1 + \frac{r}{n})^{nt}$$

A = Amount of money possessed after t years. P = Initial amount of money (i.e. principal), r = Yearly interest rate, also called annual percentage rate (APR). n = Number of times compounded per year.

① If \$7,592 is deposited in a bank account paying a 3% annual interest rate compounded monthly, how much money will be in the account in 2 years? Assume that there are no withdrawals or additional deposits.

$$\begin{aligned} A &= P(1 + \frac{r}{n})^{nt} = 7,592(1 + \frac{0.03}{12})^{12(2)} = 7,592(1 + 0.0025)^{12(2)} \\ &= 7,592(1.0025)^{12(2)} \\ &= 7,592(1.0025)^{24} \\ &\approx 7,592(1.061757) \\ &\approx \boxed{\$8,060.86} \end{aligned}$$

② If \$200.98 is deposited in a bank account paying a 0.8% annual interest rate compounded daily, how much money will be in the account in 5 years? Assume that there are no withdrawals or additional deposits

$$\begin{aligned} A &= P(1 + \frac{r}{n})^{nt} = 200.98(1 + \frac{0.008}{365})^{365(5)} \\ &\approx 200.98(1 + 0.0000219178)^{365(5)} \\ &\approx 200.98(1.0000219178)^{365(5)} \\ &\approx 200.98(1.0000219178)^{1,825} \\ &\approx 200.98(1.04081) \\ &\approx \boxed{\$209.18} \end{aligned}$$

★ ③ If \$21,340 is deposited in a bank account paying a 2% annual interest rate compounded annually, how much money will be in the account in 10 years? Assume that there are no withdrawals or additional deposits

$$\begin{aligned} A &= P(1 + \frac{r}{n})^{nt} = 21,340(1 + \frac{0.02}{1})^{(1)(10)} = 21,340(1 + 0.02)^{(1)(10)} \\ &= 21,340(1.02)^{(1)(10)} \\ &= 21,340(1.02)^{10} \\ &\approx 21,340(1.2189944) \\ &\approx \boxed{\$26,013.34} \end{aligned}$$

- ★ ④ If \$100 is deposited in a bank account paying a 5.5% annual interest rate compounded quarterly, how much money will be in the account in 20 years? Assume that there are no withdrawals or additional deposits.

$$\begin{aligned}
 A &= P\left(1 + \frac{r}{n}\right)^{nt} = 100\left(1 + \frac{0.055}{4}\right)^{4(20)} = 100(1.01375)^{4(20)} \\
 &= 100(1.01375)^{80} \\
 &= 100(1.01375)^{80} \\
 &\approx 100(2.981737) \\
 &\approx \boxed{\$298.17}
 \end{aligned}$$

- ⑤ Money is deposited in an account paying a 1% annual interest rate compounded biannually. After 7 years, the investment grows to \$3,048.16. How much money was initially deposited in the account? Assume that there were no additional deposits or withdrawals.

$$\begin{aligned}
 A &= P\left(1 + \frac{r}{n}\right)^{nt} \\
 3,048.16 &= P\left(1 + \frac{0.01}{2}\right)^{2(7)} \\
 3,048.16 &= P(1.005)^{2(7)} \\
 3,048.16 &= P(1.005)^{14} \\
 3,048.16 &= P(1.005)^{14} \\
 3,048.16 &\approx P(1.07232)
 \end{aligned}$$

$$\frac{3,048.16}{1.07232} \approx \frac{P(1.07232)}{1.07232}$$

$$P \approx \boxed{\$2,842.58}$$

- ⑥ Money is deposited in an account paying a 0.9% annual interest rate compounded monthly. After 12 years, the investment grows to \$10,076. How much money was initially deposited in the account? Assume that there were no additional deposits or withdrawals.

$$\begin{aligned}
 A &= P\left(1 + \frac{r}{n}\right)^{nt} \\
 10,076 &= P\left(1 + \frac{0.009}{12}\right)^{12(12)} \\
 10,076 &= P(1.00075)^{12(12)} \\
 10,076 &= P(1.00075)^{144} \\
 10,076 &\approx P(1.1140026)
 \end{aligned}$$

$$\frac{10,076}{1.1140026} \approx \frac{P(1.1140026)}{1.1140026}$$

$$P \approx \boxed{\$9,044.86}$$

- ⑦ Joe loans his brother \$300 at an interest rate compounded daily. If Joe's brother owes him \$360 after one year, what annual interest rate did Joe charge his brother? Assume that his brother did not pay back any of the money during the year.

$$\begin{aligned}
 A &= P\left(1 + \frac{r}{n}\right)^{nt} \\
 360 &= 300\left(1 + \frac{r}{365}\right)^{365(1)} \\
 360 &= 300\left(1 + \frac{r}{365}\right)^{365} \\
 \frac{360}{300} &= \frac{300\left(1 + \frac{r}{365}\right)^{365}}{300} \\
 1.2 &= \left(1 + \frac{r}{365}\right)^{365}
 \end{aligned}$$

$$\sqrt[365]{1.2} = \sqrt[365]{\left(1 + \frac{r}{365}\right)^{365}}$$

$$1.0004996 \approx 1 + \frac{r}{365}$$

$$0.0004996 \approx \frac{r}{365}$$

$$0.182367 \approx r$$

$$\boxed{18.24\%}$$

- ★ ⑧ Lisa received a student loan. If she does not make any payments on the loan for 18 years, and the debt grows from \$6,000 to \$32,468.11, what annual interest rate did the creditor charge if the interest is compounded semiannually?

$$A = P(1 + \frac{r}{n})^{nt}$$

$$32,468.11 = 6,000(1 + \frac{r}{2})^{2(18)}$$

$$32,468.11 = 6,000(1 + \frac{r}{2})^{36}$$

$$\frac{32,468.11}{6,000} = \frac{6,000(1 + \frac{r}{2})^{36}}{6,000}$$

$$5.4113 \approx (1 + \frac{r}{2})^{36}$$

$$\sqrt[36]{5.4113} \approx \sqrt[36]{(1 + \frac{r}{2})^{36}}$$

$$1.04802 \approx 1 + \frac{r}{2}$$

$$0.04802 \approx \frac{r}{2}$$

$$0.09604 = r$$

$$\boxed{9.6\%}$$

- ⑨ Jeff and Abigail take out a mortgage on a house. The interest is compounded quarterly at a 5% annual rate. If they do not make payments for an initial period, and the debt grows from \$436,208 to \$439,118, how long was this initial period that they did not make payments?

$$A = P(1 + \frac{r}{n})^{nt}$$

$$439,118 = 436,208(1 + \frac{0.05}{4})^{4t}$$

$$439,118 = 436,208(1 + 0.0125)^{4t}$$

$$439,118 = 436,208(1.0125)^{4t}$$

$$\frac{439,118}{436,208} = \frac{436,208(1.0125)^{4t}}{436,208}$$

$$1.00667 \approx 1.0125^{4t}$$

$$4t \approx \log_{1.0125} 1.00667$$

$$4t \approx 0.53523567$$

$$\boxed{t \approx 0.1338 \text{ years}}$$

- ⑩ If Mark deposits \$8,900, paying an annual interest rate of 2.2% compounded yearly, and his investment grows to \$11,401, how much time passed? Assume that he made no withdrawals or additional deposits.

$$A = P(1 + \frac{r}{n})^{nt}$$

$$11,401 = 8,900(1 + \frac{0.022}{1})^{(1)t}$$

$$11,401 = 8,900(1 + 0.022)^t$$

$$11,401 = 8,900(1.022)^t$$

$$\frac{11,401}{8,900} = \frac{8,900(1.022)^t}{8,900}$$

$$1.281011 \approx 1.022^t$$

$$t \approx \log_{1.022} 1.281011$$

$$\boxed{t \approx 11.38 \text{ years}}$$

Continuous Compounding

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = P\left[\left(1 + \frac{r}{n}\right)^n\right]^t = P\left[\left(1 + \frac{1}{\frac{n}{r}}\right)^n\right]^t = P\left[\left(1 + \frac{1}{\frac{n}{r}}\right)^{\frac{n}{r}}\right]^t$$

Exponent Rule V

Let n approach infinity.
As a result,
 $\left(1 + \frac{1}{\frac{n}{r}}\right)^{\frac{n}{r}}$

will approach 2.71828...

Therefore, if n approaches infinity, then

$$A = P\left[2.71828\ldots\right]^t$$

Exponent Rule V

$$A = P\{2.71828\ldots\}^{rt}$$

We give this number a special letter designation.

$$A = Pe^{rt}$$

Continuous Compounding Formula

If n becomes very large (i.e. approaches infinity) in the compound interest formula, then the equation converts to the form below, where e is Euler's number (i.e. $\sim 2.718\ldots$).

$$A = Pe^{rt}$$

⑪ If interest is compounded continuously for 2 years at a 3% rate, how much would a \$4,000 investment grow to?

$$A = Pe^{rt} = 4000e^{0.03(2)} = 4000e^{0.06} \approx 4000(1.06183) \approx \boxed{\$4,247.35}$$

⑫ If a 0.07% interest rate is compounded continuously for 10 years, and the initial investment is \$541, how much would the investment be worth?

$$A = Pe^{rt} = 541e^{0.0007(10)} = 541e^{0.007} \approx 541(1.00702) \approx \boxed{\$544.80}$$

⑬ If a 1% interest rate is compounded continuously for 8 years, and the initial investment is \$947, how much would the investment be worth?

$$A = Pe^{rt} = 947e^{0.01(8)} = 947e^{0.08} \approx 947(1.083287) \approx \boxed{\$1,025.87}$$

⑭ If a 4.23% interest rate is compounded continuously for 15 years, and the initial investment is \$7,046, how much would the investment be worth?

$$A = Pe^{rt} = 7046e^{0.0423(15)} = 7046e^{0.6345} \approx 7046(1.88607) \approx \boxed{\$13,289.31}$$

⑮ If an \$18,293 investment grows to \$40,368 in 9 years, what interest rate was applied if it was compounded continuously?

$$A = Pe^{rt}$$

$$40,368 = 18,293e^{r(9)}$$

$$40,368 = 18,293e^{9r}$$

$$\frac{40,368}{18,293} = \frac{18,293e^{9r}}{18,293}$$

$$2.2067 \approx e^{9r}$$

$$9r \approx \log_e 2.2067$$

$$9r \approx \ln 2.2067$$

$$9r \approx 0.7915189$$

$$r \approx 0.08794$$

$$\boxed{8.79\%}$$

⑯ If an investment grows from \$5,023.14 to \$6,112.14 at a 3.7% annual interest rate compounded continuously, how much time passed during this growth?

$$A = Pe^{rt}$$

$$6,112.14 = 5,023.14e^{0.037t}$$

$$\frac{6,112.14}{5,023.14} = \frac{5,023.14e^{0.037t}}{5,023.14}$$

$$1.2167966 \approx e^{0.037t}$$

$$0.037t \approx \log_e 1.2167966$$

$$0.037t \approx \ln 1.2167966$$

$$0.037t \approx 0.19622$$

$$\boxed{t \approx 5.3 \text{ years}}$$

⑰ If an \$800.99 investment grows to \$31,024 in 30 years, what interest rate was applied if it was compounded continuously?

$$A = Pe^{rt}$$

$$31,024 = 800.99e^{r(30)}$$

$$31,024 = 800.99e^{30r}$$

$$\frac{31,024}{800.99} = \frac{800.99e^{30r}}{800.99}$$

$$38.732 \approx e^{30r}$$

$$30r \approx \log_e 38.732$$

$$30r \approx \ln 38.732$$

$$30r \approx 3.656667$$

$$r \approx 0.12188$$

$$\boxed{12.19\%}$$

⑱ If an investment grows from \$126,476 to \$130,000 at a 4% annual interest rate compounded continuously, how much time passed during this growth?

$$A = Pe^{rt}$$

$$130,000 = 126,476e^{0.04t}$$

$$\frac{130,000}{126,476} = \frac{126,476e^{0.04t}}{126,476}$$

$$1.02786 \approx e^{0.04t}$$

$$0.04t \approx \log_e 1.02786$$

$$0.04t \approx \ln 1.02786$$

$$0.04t \approx 0.02748$$

$$\boxed{t \approx 0.687 \text{ years}}$$